



UNIVERSITI MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

CSE3433

Software Architecture

Cloud-Native Learning Management System (LMS)
(Cloud-Native / Layered Architecture)

Prepared by:

Group No.5

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COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOUR
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WEEKLY LOG ENTRIES

PHASE 1: Project Proposal

WEEK 5

Undertaken Tasks & Weekly Activities	● Successfully established a comprehensive development lifecycle roadmap which provide clear direction for all team members. ● Performed a complete review of different platforms to deploy cloud services so that the architecture can meet future scalability and performance needs.
Individual Contributions (as Project Manager):	● Selected the complete set of technology stack and, especially, chose Railway as the environment for the application. ● Established Apache NetBeans as the standard local development environment for the entire team.
Challenges Encountered	● Felt very much difficult to represent the selected multiple platform cloud architecture in terms of the normal architectural grading criteria as taught in the course. ● Also found complexities to reconcile the definitions of standard layered architecture with the requirements of microservices architecture.
Action Plans	● Developed documentation on the strict separation of application layer on Railway and data layer on AWS. ● All documents will prove that the design satisfies the principle of "isolation of concerns."

Table 1 : Weekly Progress Log (Week 5)

PHASE 2: Architecture Design

WEEK 6

Undertaken Tasks & Weekly Activities	● Worked on the definition of key Learning Management System (LMS) modules needed to achieve the functionality of the platform.
Individual Contributions (as Project Manager):	● Conducted strategic workspace organization to make the development process more effective and avoid versioning problems.
Challenges Encountered	● Faced synchronization issues while trying to connect several local environments at once. ● Faced major challenges with configuring one common live AWS cloud database for the whole team.
Action Plans	● Established the common environmental variables standard to make configuration of all project settings uniform. ● Took care of the implementation of secure cloud connection strings to avoid configuration conflicts among the different local development environments.

Table 2 : Weekly Progress Log (Week 6)

WEEK 7

Undertaken Tasks & Weekly Activities	● Completed comprehensive system context and component diagram to be included in the project report. ● Completed complex interaction and deployment diagrams that were necessary for the submission of Deliverable 2.
Individual Contributions (as Project Manager):	● Validated the interaction diagrams in order to guarantee their accuracy concerning the system data flow. ● Confirmed that all the user interface interactions properly invoked corresponding backend services through the API gateway.
Challenges Encountered	● Failed to properly depict modern multi-provider cloud infrastructure using conventional UML nodes. ● Observed considerable shortcomings in standard documentation tools in representing particular infrastructure elements of the cloud environment.
Action Plans	● Designed Railway as the main application runtime node to differentiate it from other infrastructure nodes. Depicted AWS as an independent infrastructure block interconnected with the architecture.

Table 3 : Weekly Progress Log (Week 7)

PHASE 3: Project Observation 1 & Prototype Progress

WEEK 8

Undertaken Tasks & Weekly Activities	● Compiled all technical documents and presentation material required for the coming Project Observation 1. ● Began the creation and design process of the required frontend features in the Apache NetBeans IDE.
Individual Contributions (as Project Manager):	● Organized all the application resources in order to make sure that it is ready for deployment during the observation. ● Did the first prototype to make sure that the basic user interface features are working properly.
Challenges Encountered	● Faced high level of technical complexity in designing the real-time notification synchronization function. ● Experiencing problems with inconsistency of data between the Railway application instances and AWS database.
Action Plans	● The design of highly optimized API endpoints which will deal with the large number of data requests from the frontend. ● Implementation of AWS PostgreSQL database triggers in order to update the data efficiently.

Table 4 : Weekly Progress Log (Week 8)

PHASE 3: Project Observation 1 & Prototype Progress

Tool Used	ChatGPT
Purpose	Brainstorming how real-time components work across distributed cloud hosting environments.
Prompt Use	"How do I explain real-time data synchronization features between a web service hosted on Railway and a cloud database running on AWS?"
Output Summary	Explained how structured API polling, persistent WebSocket connections and managed database listener triggers handle live updates efficiently without heavy performance overhead.
How it used	Used this information to help explain our real time notification synchronization logic during our team planning session.

Table 5 : GenAI Usage Record (Week 8)

WEEK 9

Undertaken Tasks & Weekly Activities	● Discussed project architecture and design at Project Observation 1 with the lecturer. ● Discussed selected approach of deployment and its benefits to the academic evaluators.
Individual Contributions (as Project Manager):	Considered the necessary inputs from the evaluators to improve user flow in the system. Improved the quality of the user interface to improve usability of the system.
Challenges Encountered	Recognized the need for complete isolation of user authentication data from the other course management data. Recognized security threats present in the public part of the Learning Management System.
Action Plans	Set up different and secure database schemas in AWS PostgreSQL. Used stronger security measures to store user credentials.

Table 6 : Weekly Progress Log (Week 9)

Tool Used	Gemini
Purpose	Formatting presentation points based on feedback.
Prompt Use	"How to structure database user isolation strategies between application services and AWS RDS PostgreSQL after an architectural review feedback."
Output Summary	Recommended using strict PostgreSQL role privileges, separate database schemas and keeping application auth tokens secure.
How it used	Used this layout approach to plan the database security adjustments with Aida

Table 7: GenAI Usage Record (Week 9)

WEEK 10

Undertaken Tasks & Weekly Activities	● Ongoing monitoring of the prototype status for proper alignment of the project development with the plan. ● Successful integration of the working login screen into the Railway cloud environment.
Individual Contributions (as Project Manager):	● Finalizing of the login page in order to provide secure user access to the platform. ● Fixing of data type inconsistencies between the NetBeans inputs and AWS PostgreSQL database fields.
Challenges Encountered	● Determined significant inconsistency between the local application and cloud host database fields. ● Experiencing validation problems during data type mapping.
Action Plans	Creating detailed guidelines for data type matching. Achieving full consistency between the local and cloud host environments.

Table 8 : Weekly Progress Log (Week 10)

Tool Used	Gemini
Purpose	Resolving data-type mapping inconsistencies between frontend and cloud database.
Prompt Use	"What is the standard matching data type in PostgreSQL on AWS RDS for a basic string array coming from an application built in NetBeans?"
Output Summary	Recommended using the standard text [] array type or JSONB for flexible object schemas.
How it used	Used this technical guidance to adjust frontend input validation and database column properties.

Table 9: GenAI Usage Record (Week 10)

WEEK 11

Undertaken Tasks & Weekly Activities	● We ran our final integration tests on the prototype and successfully demonstrated our initial working system for Deliverable 3.
Individual Contributions (as Project Manager):	● Collected user interface screenshots for the full system functionality. ● Collected all the needed GitHub repository links to complete the submission.
Challenges Encountered	● Faced deployment issues on Railway in the end while preparing the submission. ● Found out that connection strings were missing in the project console.
Action Plans	● Updated the Railway dashboard by including all the missing connection strings to bring back the functionality. ● Redeployed the application to make the system accessible again.

Table 10 : Weekly Progress Log (Week 11)

Tool Used	Gemini
Purpose	Creating clear technical documentation for a project submission.
Prompt Use	"Give me a clean layout format to present multi-provider cloud deployment screenshots (Railway and AWS) and repository links for an academic system prototype report."
Output Summary	Provided a neat template with a clear description block, system environment notes, and source links.
How it used	Used this to format our Deliverable 3 submission document.

Table 11: GenAI Usage Record (Week 11)

PHASE 4: Project Observation

WEEK 12

Undertaken Tasks & Weekly Activities	● We presented our integrated system for Project Observation 2, demonstrating how our live application layers deployed on Railway communicate with our AWS cloud infrastructure resources.
Individual Contributions (as Project Manager):	● Successfully showcased the responsive nature of the front-end interface in different devices. ● Fixed the performance issues related to database request latency when it was high.
Challenges Encountered	● There were observed issues related to network latency during the time of peak database requests from the front-end. ● There were observed performance issues of users during high traffic.
Action Plans	● The issue of connection pooling was fixed. The server environment variables were optimized for stable connections.

Table 12 : Weekly Progress Log (Week 12)

Tool Used	ChatGPT
Purpose	Troubleshooting network latency issues between the frontend and cloud database.
Prompt Use	"How do I minimize database connection latency between an application hosted on Railway and a PostgreSQL database running on AWS RDS?"
Output Summary	Suggested enabling connection pooling, optimizing foreign key indexing, and ensuring geographic alignment of cloud regions.
How it used	Applied these optimization strategies to the frontend connection string configuration.

Table 13: GenAI Usage Record (Week 12)

WEEK 13

Undertaken Tasks & Weekly Activities	● Collected all technical information needed to be mentioned in the final project report. Collected all diagrams and recorded all the environments that will be used in project submission.
Individual Contributions (as Project Manager):	● Styled the Technology Stack section and Deployment section to make them readable for our readers. ● Mentioned all the frontend integration requirements and dependencies.
Challenges Encountered	● Felt difficulties in integrating the frontend deployment notes with the very technical backend security notes written by other members of our team. ● Found discrepancies in the tone used in various sections of the report.
Action Plans	● Improved the transitions within the report to maintain a structured flow in the report. ● Improved the final report to keep the tone of the whole report very professional and academic.

Table 14 : Weekly Progress Log (Week 13)

Tool Used	Gemini
Purpose	Structuring the final project report with a clear academic layout.
Prompt Use	"How do I neatly structure an implementation and deployment summary section for a university report when using multiple cloud providers like Railway and AWS?"
Output Summary	Recommended using a clear tabular breakdown or distinct sub headers dividing the compute/hosting layer from the data persistence layer.
How it used	Used formatting suggestion to organize our report Technology Stack and Deployment sections cleanly.

Table 15: GenAI Usage Record (Week 13)